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Static and free vibration analysis of sandwich cylindrical shell based on theory of elasticity and using DQM

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Abstract In this paper, static and free vibration analysis of a sandwich cylindrical shell is performed using theory of elasticity formulation. The core layer is made of functionally graded material with material properties varying along the thickness direction according to a simple power law. For the case of simply supported boundary conditions, equations of motion and equilibrium equations are solved analytically by applying a state-space technique along the radial direction and Fourier series expansion along the axial and circumferential direction. When boundary conditions are not simply supported, a semi-analytical solution is performed by using the differential quadrature method along the axial direction. The present approach is validated by comparing the obtained numerical results with those published in the available literature. Moreover, effects of boundary conditions, graded direction, mid-radius to thickness and length to mid-radius ratios on bending and vibration behavior are considered.

1 Introduction

Sandwich structures are used widely in many industrial applications due to their high strength and stiffness to weight ratios. Using an FGM core layer in sandwich structures causes to increase resistance of debonding failure which is due to the different stiffness between the face sheets and the core layer. Many studies have been accomplished on static and vibration behavior of sandwich structures in recent years. Varadan and Bhaskar [1] presented a three-dimensional solution for an orthotropic cylindrical shell with simply supported boundary condition. Kabir and Askar [2] carried out free vibration analysis of cylindrical panels with higher-order shear deformation theory. Tutuncu [3] investigated bending behavior of FGM cylindrical shells with exponentially varying properties using elasticity theory as well as Frobenius series. Alibeigloo [4] investigated static and free vibration axisymmetric angle-ply laminated cylindrical shells using the state-space differential quadrature method. Zhao et al. [5] carried out thermoelastic and vibration analysis of functionally graded cylindrical shells. Based on a 2D higher-order deformation theory, Matsunaga [6] studied free vibration and stability of functionally graded circular cylindrical shells. Based on Carrera's unified formulation and the principle of virtual displacements, Brischetto [7] investigated thermal stress of thick and thin multilayered cylindrical and spherical shells including carbon fiber-reinforced layers and/or a central soft core. Pradyumna and Bandyopadhyay [8] studied free vibration and buckling of functionally graded shell panels in thermal environments using higher-order shear deformation theory. Based on theory of elasticity, Alibeigloo and Nouri [9] carried out static analysis of a functionally graded cylindrical shell with piezoelectric layers using DQM. Piezoelectricity analysis of an FGM shell embedded in piezoelectric layers was accomplished by Alibeigloo and

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Chen [10] using elasticity theory. Vibration and buckling analysis of two-layered functionally graded cylindrical shells by considering the effects of transverse shear and rotary inertia was investigated by Sepiani et al. [11]. Malekzadeh and Heydarpour [12] studied transient thermoelastic behavior of FGM cylindrical shells subjected to moving thermo-mechanical loads by using the finite element method. Rahmani et al. [13] utilized high-order sandwich panel theory to investigate free vibration behavior of circular cylindrical composite sandwich shells subjected to external loads. Alibeigloo et al. [14] studied free vibration of an FGM cylindrical shell bonded to thin piezoelectric layers using theory of elasticity. Based on higher-order shear deformation theory with von-Karman nonlinearity using modified C0 continuity, Lal et al. [15] discussed the effect of random system properties on the thermal post-buckling temperature of a laminated composite cylindrical shell panel. Alipour and Shariyat [16] used a Maclaurin-type series solution and an elasticity-equilibrium-based zigzag theory to analyze bending behavior of FGM circular sandwich plates. Based on the principle of virtual displacements, Wu and Kuo [17] used a unified formulation of a finite cylindrical layer method to study bending and free vibration of simply supported FGM sandwich circular hollow cylinders. Sofiyev and Kuruoglu [18] studied buckling and free vibration of shear deformable FGM orthotropic cylindrical shells under external pressure. Alibeigloo and Liew [19] carried out free vibration analysis of a sandwich cylindrical panel with FGM core using three-dimensional theory of elasticity. Sun et al. [20] used a homogenization method to carry out static analysis of multilayered hollow anisotropic cylindrical structures. Free vibration analysis of an axisymmetric FGM thermoelastic cylindrical shell was carried out by Sharma et al. [21] using the Laplace transform technique. Plagianakos and Papadopoulos [22] used a higher-order layer-wise theory as well as the finite element method to predict global piezoelectric response, local piezoelectric variables across the thickness and inter-laminar shear stress at the interface of adjacent layers. Sburlati and Kashtalyan [23] performed a closed form solution for the static behavior of sandwich pipes with two thin FGM interlayers between the core layer and inner/outer pipes using theory of elasticity. A thermoelastic solution for an FGM or multilayer shell was presented by Kugler et al. [24] using an efficient low-order finite shell element. Based on Midlin's theory, Thomas and Roy [25] analyzed vibration of an FG-CNTRC shell using the finite element method. Norouzi and Alibeigloo [26] used the state-space differential quadrature method to investigate the bending behavior of an FGM viscoelastic cylindrical panel subjected to radial uniform pressure. To the authors' knowledge, however, static and free vibration analysis of sandwich cylindrical shells with FGM core and various boundary conditions has not been reported yet. In the present work, we use the governing differential equation and constitutive relations to derive state-space equations along the radial coordinate. For a sandwich shell with simply supported edges condition, the state-space differential equations can be solved analytically by using Fourier series expansion along the radial and circumferential coordinates, whereas for support other than simply, they can be solved semi-analytically by using DQM.

2 Problem description

A sandwich cylindrical shell with length L , total thickness h , inner and outer radius r_i and r_o , Fig. 1, is considered (Fig. 1). The core layer is made by FGM and the inner and outer facing sheets are metal and ceramic, respectively.

The Poisson's ratio of the FGM core layer is considered to be constant while the Young's modulus and material density are assumed to vary continuously through-the-thickness according to the following power-law relation:

$$E = E_0 \left(\frac{r}{r_i + h_m} \right)^{m_1}, \quad \rho = \rho_0 \left(\frac{r}{r_i + h_m} \right)^{m_2} \quad r_i + h_m \leq r \leq r_o - h_c. \quad (1)$$

Here $m_1 = \frac{\ln \frac{E_h}{E_0}}{\ln \frac{r_o - h_c}{r_i + h_m}}$, $m_2 = \frac{\ln \frac{\rho_h}{\rho_0}}{\ln \frac{r_o - h_c}{r_i + h_m}}$ and E_0, ρ_0, E_h, ρ_h are Young's modulus and material density at the inner and outer surfaces of the FGM core layer, respectively; h_m, h_c are metal and ceramic layer thickness.

Equations of motion, in absence of body forces, in cylindrical coordinates are:

$$\begin{aligned} \sigma_{r,r} + \frac{1}{r} \tau_{r\theta,\theta} + \tau_{rz,z} + \frac{1}{r} (\sigma_r - \sigma_\theta) &= \rho \frac{\partial^2 u_r}{\partial t^2}, \\ \tau_{r\theta,r} + \frac{1}{r} \sigma_{\theta,\theta} + \tau_{z\theta,z} + \frac{2}{r} \tau_{r\theta} &= \rho \frac{\partial^2 u_\theta}{\partial t^2}, \\ \tau_{rz,r} + \frac{1}{r} \tau_{z\theta,\theta} + \sigma_{z,z} + \frac{1}{r} \tau_{rz} &= \rho \frac{\partial^2 u_z}{\partial t^2}, \end{aligned} \quad (2)$$

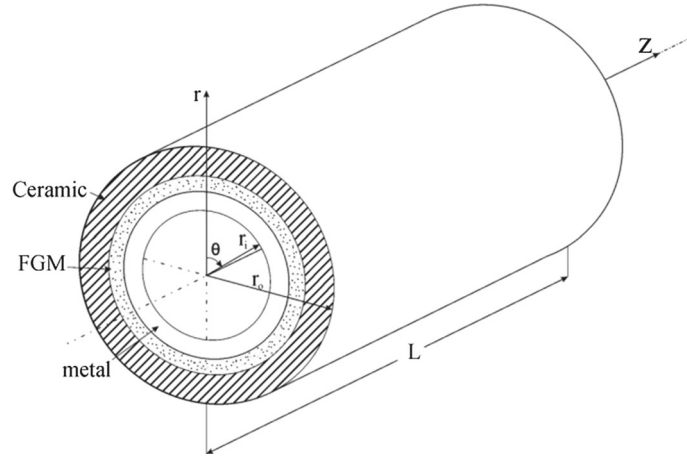


Fig. 1 Geometry of FGM sandwich cylindrical shell

where σ_i, τ_{ij} ($i = r, \theta, z$) are normal and shear stresses; u_i ($i = r, \theta, z$) are displacement components along the radial, circumferential and axial direction; ρ is the material density, respectively.

The stress–displacement relations for the core layer in linear elasticity are

$$\begin{aligned}\sigma_r &= \frac{E}{(1+\nu)(1-2\nu)} \left[(1-\nu)u_{r,r} + \frac{\nu}{r}(u_{\theta,\theta} + u_r) + \nu u_{z,z} \right], \\ \sigma_\theta &= \frac{E}{(1+\nu)(1-2\nu)} \left[\nu u_{r,r} + \frac{1-\nu}{r}(u_{\theta,\theta} + u_r) + \nu u_{z,z} \right], \\ \sigma_z &= \frac{E}{(1+\nu)(1-2\nu)} \left[\nu u_{r,r} + \frac{\nu}{r}(u_{\theta,\theta} + u_r) + (1-\nu)u_{z,z} \right], \\ \tau_{z\theta} &= \frac{E}{2(1+\nu)} \left(u_{\theta,z} + \frac{1}{r}u_{z,\theta} \right) \quad \tau_{zr} = \frac{E}{2(1+\nu)} (u_{z,r} + u_{r,z}), \quad \tau_{r\theta} = \frac{E}{2(1+\nu)} \left(\frac{1}{r}u_{r,\theta} + u_{\theta,r} - \frac{u_\theta}{r} \right).\end{aligned}\quad (3)$$

The state-space equations for the FGM core layer are derived by using Eqs. (2) and (3):

$$\begin{aligned}\sigma_{r,r} &= \left(-\frac{1}{r} + \frac{\nu}{r(1-\nu)} \right) \sigma_r + \frac{E\nu}{r(1-\nu^2)} u_{z,z} + \frac{E}{r^2(1-\nu^2)} u_{\theta,\theta} \\ &\quad + \left(\frac{E}{r^2(1-\nu^2)} + \rho \frac{\partial^2}{\partial t^2} \right) u_r - \tau_{r,z} - \frac{1}{r} \tau_{r\theta,\theta} u_{z,r} = \frac{2(1+\nu)}{E} \tau_{r,z} - u_{r,z}, \\ u_{\theta,r} &= \frac{2(1+\nu)}{E} \tau_{r\theta} + \frac{u_\theta}{r} - \frac{1}{r} u_{r,\theta}, \\ u_{r,r} &= \frac{(1+\nu)(1-2\nu)}{(1-\nu)E} \sigma_r - \frac{\nu}{1-\nu} u_{z,z} - \frac{\nu}{r(1-\nu)} u_{\theta,\theta} - \frac{\nu}{r(1-\nu)} u_r, \\ \tau_{r,z,r} &= -\frac{\nu}{1-\nu} \sigma_{r,z} - \left(\frac{E}{1-\nu^2} u_{z,z,z} - \rho \frac{\partial^2 u_z}{\partial t^2} \right) - \frac{\nu^2}{r(1-\nu)} u_{\theta,\theta} - \frac{E\nu}{r(1-\nu^2)} u_{r,z} - \frac{\tau_{r,z}}{r}, \\ \tau_{r\theta,r} &= -\frac{E}{2(1+\nu)} u_{\theta,z,z} + \rho \frac{\partial^2 u_\theta}{\partial t^2} - \frac{2\tau_{r\theta}}{r}.\end{aligned}\quad (4)$$

Equation (4) can be expressed in matrix form as

$$\frac{d\delta_f}{dr} = G_f \delta_f, \quad (5)$$

where $\delta_f = \{ \sigma_r \ u_z \ u_\theta \ u_r \ \tau_{r,z} \ \tau_{z\theta} \}^T$, and G_f is a coefficient matrix (see Appendix I).

Similarly, the state-space equations for the ceramic and metal layers can be derived by consideration of constant Young's modulus and material density:

$$\frac{d\delta_m}{dr} = G_m \delta_m, \quad (6a)$$

$$\frac{d\delta_c}{dr} = G_c \delta_c, \quad (6b)$$

where G_m and G_c are coefficient matrices for metal and ceramic facing sheets, respectively.

The in-plane stresses σ_z , σ_θ , $\tau_{z\theta}$ for each layer in terms of induced variables are as follows:

$$\begin{aligned} \sigma_z &= \frac{v}{1-v} \sigma_r + \frac{E}{1-v^2} u_{z,z} + \frac{Ev}{r(1-v^2)} u_{\theta,\theta} + \frac{Ev}{r(1-v^2)} u_r, \\ \sigma_\theta &= \frac{v}{1-v} \sigma_r + \frac{Ev}{1-v^2} u_{z,z} + \frac{E}{r(1-v^2)} u_{\theta,\theta} + \frac{E}{r(1-v^2)} u_r, \\ \tau_{z\theta} &= \frac{E}{2(1+v)} \left(u_{\theta,z} + \frac{1}{r} u_{z,\theta} \right). \end{aligned} \quad (7)$$

In this study, the effect of combinations of different boundary conditions such as simply (S) support, clamped (C) support and free (F) support at the edges $z = 0, L$ on bending and free vibration of the sandwich shell are considered:

$$S: \sigma_z = u_r = u_\theta = 0 \quad C: u_r = u_\theta = u_z = 0 \quad F: \sigma_z = \tau_{rz} = \tau_{z\theta} = 0. \quad (8)$$

3 Exact solution

Fourier series expansion of the displacements and stresses according to the following satisfies the simply-supported boundary conditions:

$$\begin{aligned} u_r &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} U_r(r) \sin(\beta_m \theta) \sin(p_n z) e^{i\omega t} & u_\theta &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} U_\theta(r) \cos(\beta_m \theta) \sin(p_n z) e^{i\omega t} \\ u_z &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} U_z(r) \sin(\beta_m \theta) \cos(p_n z) e^{i\omega t} & \sigma_r &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \sigma'_r(r) \sin(\beta_m \theta) \sin(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} \\ \sigma_\theta &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \sigma'_\theta(r) \sin(\beta_m \theta) \sin(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} & \sigma_z &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \sigma'_z(r) \sin(\beta_m \theta) \sin(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} \\ \tau_{rz} &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \tau'_{rz}(r) \sin(\beta_m \theta) \cos(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} & \tau_{r\theta} &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \tau'_{r\theta}(r) \cos(\beta_m \theta) \sin(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} \\ \tau_{z\theta} &= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \tau'_{z\theta}(r) \cos(\beta_m \theta) \cos(p_n z) \left(\frac{r}{r_i + h_m} \right)^{m_1} e^{i\omega t} \end{aligned} \quad (9)$$

Here $p_n = \frac{n\pi}{L}$, $\beta_m = m = 1, 2, 3, \dots$, and ω is the natural frequency of the shell.

For convenience, the following non-dimensional quantities are introduced:

$$\begin{aligned} \sigma_{ij}^- &= \frac{\sigma'_{ij}}{p_0}, \quad U_i^- = \frac{U_i}{h} \quad (i = r, \theta, z), \quad h^- = \frac{h}{R_m}, \quad p_n^- = L p_n, \quad z^- = \frac{z}{L}, \quad E^- = \frac{E}{p_0}, \\ r^- &= \frac{r}{R_m}, \quad \omega^- = \omega \sqrt{\frac{R_m \rho_m h}{p_0}}, \end{aligned} \quad (10)$$

where R_m and ρ_m are the mid-radius of sandwich shell and the material density of the metal layer, respectively; p_0 is the internal uniform pressure; ω is the frequency.

Substitution of Eqs. (9) and the (10) into Eq. (5) leads to the dimensionless state-space first-order differential equations for the FGM layer as

$$\frac{d\delta_f^-}{dr^-} = G_f^- \delta_f^-, \quad (11)$$

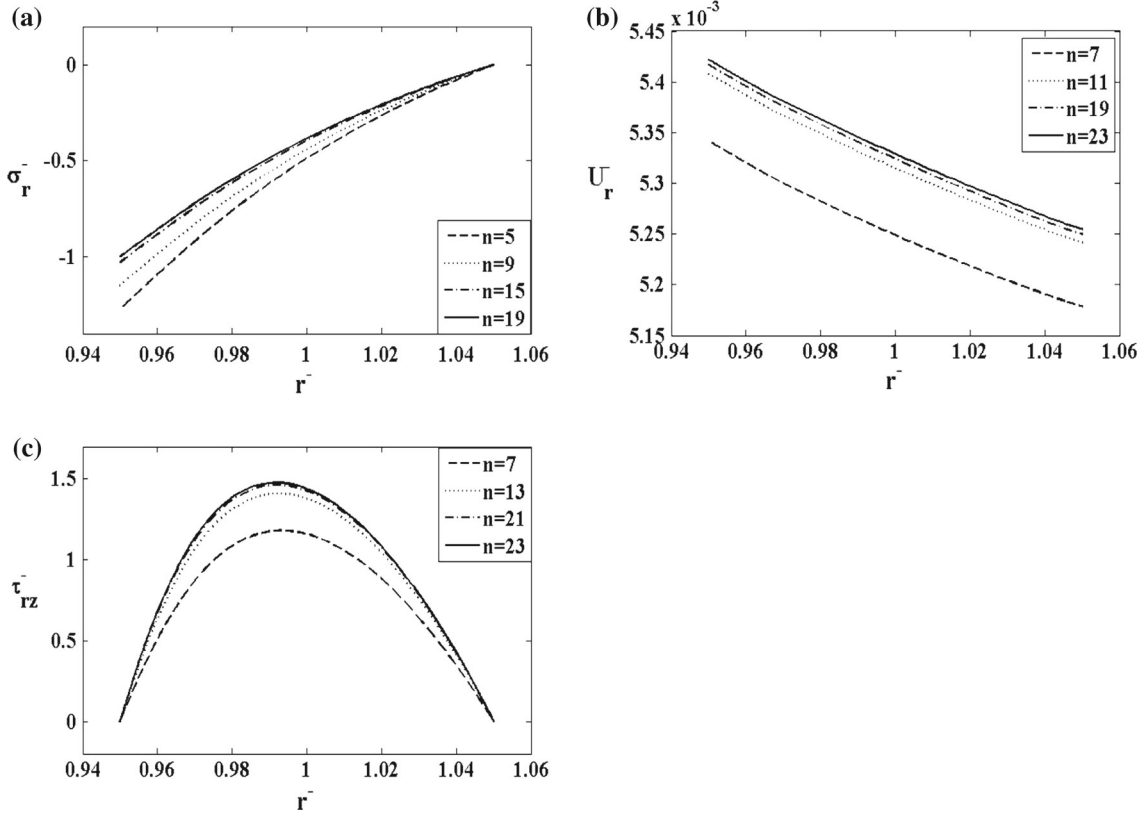


Fig. 2 Convergence of the through-the-thickness distribution of stresses and displacement of sandwich shell for $S = 10$. **a** Radial stress, **b** radial displacement, **c** transverse shear stress

where G_f^- is the coefficient matrix (see Appendix II). Since the element of G_f^- is not constant, it is difficult to solve Eq. (11) directly. In this paper, a layer-wise technique is used to divide the FGM layer into N fictitious thin layers. Therefore, we can assume G_f^- within each layer (denoted as G_{fk}^- at the mid-radius of the k th layer) to be constant. The general solution to Eq. (11) for the k th fictitious layer of the FGM is

$$\delta_{fk}^-(r^-) = \delta_{0k}^- e^{G_{fk}^-(r^- - r_{k-1}^-)} \quad r_{k-1}^- \leq r^- \leq r_k^-, \quad (12)$$

where $r_{k-1}^- = r_i^- + h_m^- + \frac{(k-1)h_f^-}{N}$, $r_k^- = r_i^- + h_m^- + \frac{kh_f^-}{N}$, $\delta_{0k}^- = \delta_{fk}^- \big|_{r=r_{k-1}^-}$.

Equation (12) at r_k^- yields

$$\delta_{fk}^-(r_k^-) = M_{fk} \delta_{0k}^-. \quad (13)$$

Here $M_{fk} = \exp(G_{fk}^- \frac{h_f^-}{N})$.

By considering the continuity condition of the state variables at each fictitious interface, a relation between the state variables at the inner and outer surfaces of the FGM core layer can be derived as

$$\delta_{fh}^- = M_f \delta_{0f}^-. \quad (14)$$

In Eq. (14), $M_f = \prod_{k=N}^1 \exp(\frac{G_{fk}^- h_f^-}{N})$, and δ_{fh}^- , δ_{0f}^- are the state vectors at the inner and outer surface of the FGM layer, respectively.

Similarly, the general solutions to Eqs. (6a) and (6b) are, respectively,

$$\delta_{mj}^-(r^-) = \delta_{0j}^- e^{G_{mj}^-(r^- - r_{j-1}^-)} \quad r_{j-1}^- \leq r^- \leq r_j^-, \quad (15a)$$

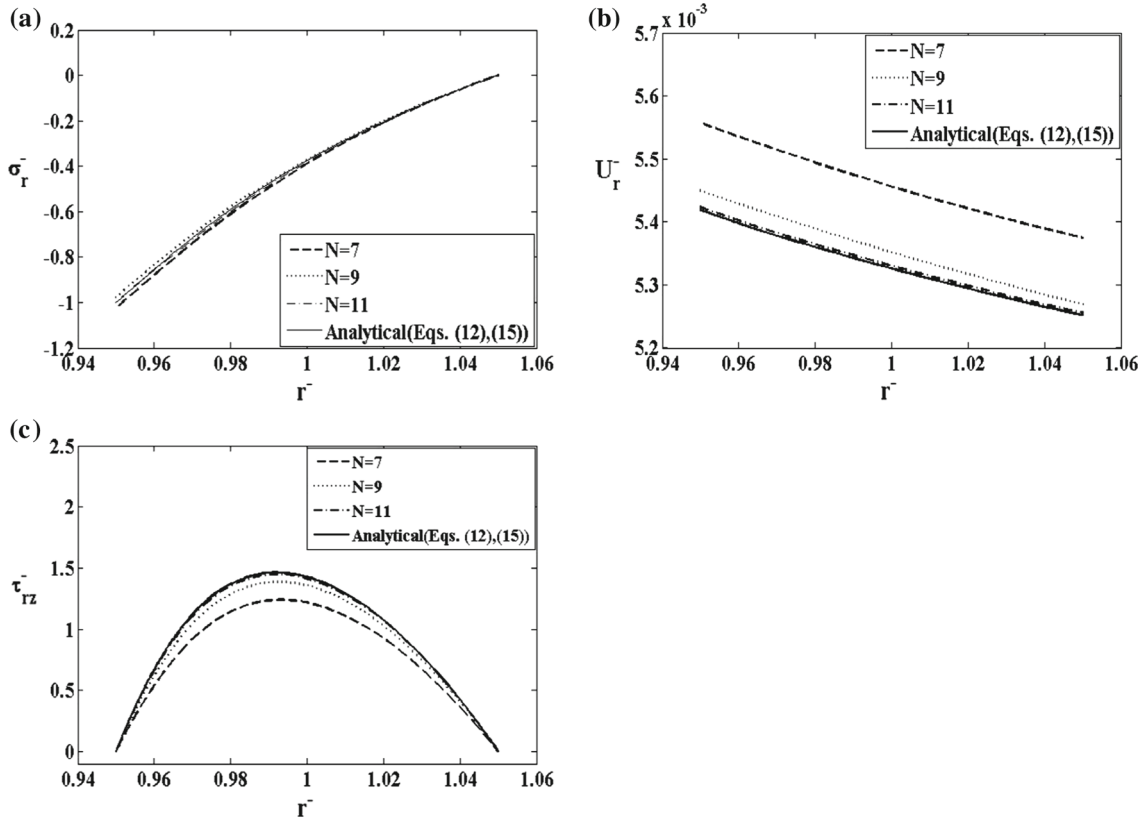


Fig. 3 Convergence of the through-the-thickness distribution of stresses and displacement of sandwich shell with SS conditions and $S = 10$. **a** Radial stress, **b** radial displacement, **c** transverse shear stress

where $r_{j-1}^- = r_i^- + \frac{(j-1)h_m^-}{N}$, $r_j^- = r_i^- + \frac{jh_m^-}{N}$, $\delta_{0j}^- = \delta_{mj}^- \Big|_{r^-=r_{j-1}^-}$,

$$\delta_{ci}^-(r^-) = \delta_{0i}^- e^{G_{ci}^-(r^- - r_{i-1}^-)} \quad r_{i-1}^- \leq r^- \leq r_i^-, \quad (15b)$$

where $r_{i-1}^- = r_o^- - h_c^- + \frac{(i-1)h_c^-}{N}$, $r_i^- = r_o^- - h_c^- + \frac{ih_c^-}{N}$, $\delta_{0i}^- = \delta_{ci}^- \Big|_{r^-=r_{i-1}^-}$.

Imposing equilibrium and continuity conditions at the fictitious interfaces of the metal and ceramic layers leads to the following relations between the state variables at the inner and outer surfaces of the metal and ceramic face sheets, respectively:

$$\delta_{mh}^- = M_m \delta_{0m}^-, \quad (16a)$$

$$\delta_{ch}^- = M_c \delta_{0c}^-, \quad (16b)$$

where $M_m = \prod_{k=N}^1 \exp\left(\frac{G_{mj}^- h_m^-}{N}\right)$, $M_c = \prod_{k=N}^1 \exp\left(\frac{G_{ci}^- h_c^-}{N}\right)$ and δ_{0m}^- , δ_{0c}^- , δ_{mh}^- , δ_{ch}^- are the state vectors at the inner and outer surfaces of the metal and ceramic layers, respectively.

The global matrix relating the state vector at the outer surface to the one at the inner surface of the FGM sandwich shell can be derived by imposing the equilibrium and continuity conditions at the face/core interfaces as

$$\delta^-(r_o^-) = A \delta^-(r_i^-). \quad (17)$$

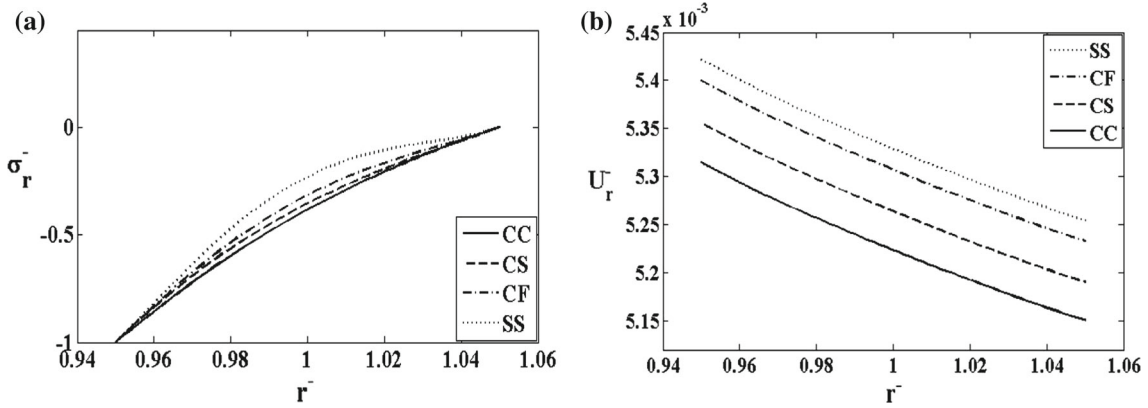
Here $A = M_c T M_f M_m$ and T is defined in Appendix III.

In the case of free vibration, the inner and outer surfaces of the FGM sandwich shell are traction free as follows:

$$\sigma_r^- = \tau_{rz}^- = \tau_{r\theta}^- = 0 \quad \text{at } r^- = r_i^- \text{ and } r_o^-. \quad (18)$$

Table 1 Comparison of dimensionless radial displacement and stress

β	Method	$\bar{U}_r \begin{pmatrix} \bar{r} = \bar{r}_i \\ \bar{r} = \bar{r}_m \\ \bar{r} = \bar{r}_o \end{pmatrix}$	$\bar{\sigma}_r \begin{pmatrix} \bar{r} = \bar{r}_i \\ \bar{r} = \bar{r}_m \\ \bar{r} = \bar{r}_o \end{pmatrix}$
2	Present analytical	0.1467	-1.0352
		0.1116	-0.4608
		0.1061	-0.0272
		0.1555	-0.9741
		0.1195	-0.4397
	Tutuncu [3]	0.1114	-0.0259
		0.2871	-1.0325
		0.2403	-0.4213
		0.2219	-0.0272
		0.3015	-0.9741
3	Present analytical	0.2499	-0.4009
		0.2336	-0.0259
		0.6073	-1.0325
		0.5099	-0.3865
		0.4750	-0.0272
	Tutuncu [3]	0.6437	-0.9741
		0.5405	-0.3664
		0.4970	-0.0259


Fig. 4 Influence of boundary condition on the dimensionless radial stress and displacement of sandwich shell with $S = 10$. **a** Radial stress, **b** radial displacement

In static analysis, it is assumed that the inner surface of the metal layer is subjected to uniform pressure, p_0 , whereas the outer surface is free from traction:

$$\sigma_r^- = \tau_{rz}^- = \tau_{r\theta}^- = 0 \quad \text{at } r^- = r_o^-, \quad (19a)$$

$$\sigma_r^- = \frac{\sigma_r}{p_0}, \quad \tau_{rz}^- = \tau_{r\theta}^- = 0 \quad \text{at } r^- = r_i^-. \quad (19b)$$

Substituting Eq. (18) into Eq. (17), the following eigenfrequency equation can be obtained:

$$|A_{ij}| = 0 \quad (i = 1, 5, 6; j = 2, 3, 4). \quad (20)$$

Natural frequencies are computed by solving Eq. (20).

In static analysis we set $\omega = 0$ and insert Eqs. (19a) and (19b) into Eq. (17), which results in the following system of non-homogeneous equations:

$$\begin{bmatrix} A_{12} & A_{13} & A_{14} \\ A_{22} & A_{23} & A_{24} \\ A_{32} & A_{33} & A_{34} \end{bmatrix} \begin{bmatrix} U_z^- \\ U_\theta^- \\ U_r^- \end{bmatrix}_{r^- = r_i^-} = \begin{bmatrix} A_{11} \\ A_{21} \\ A_{31} \end{bmatrix} p_n^-. \quad (21)$$

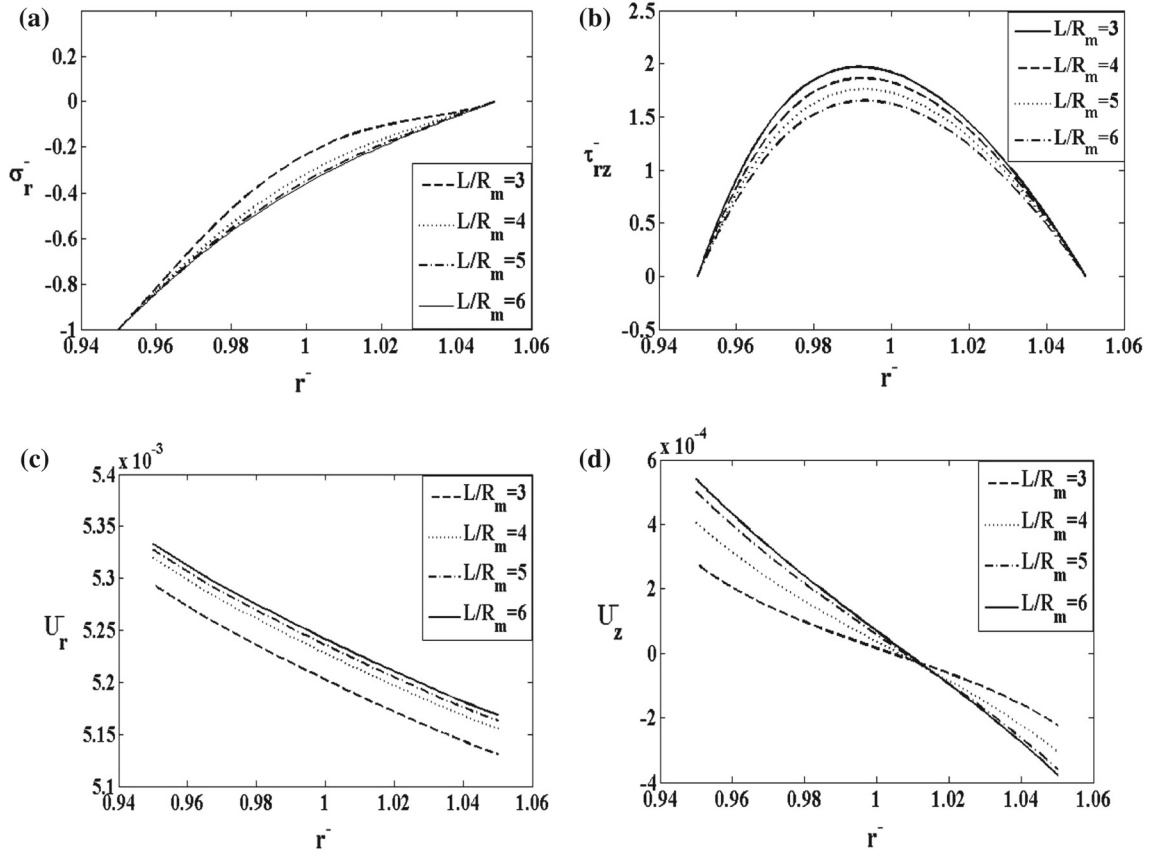


Fig. 5 Effect of length to mid-radius ratio on the non-dimensional stresses and displacements of the sandwich shell with CC boundary condition and $S = 10$. **a** Radial stress, **b** transverse shear stress, **c** radial displacement, **d** axial displacement

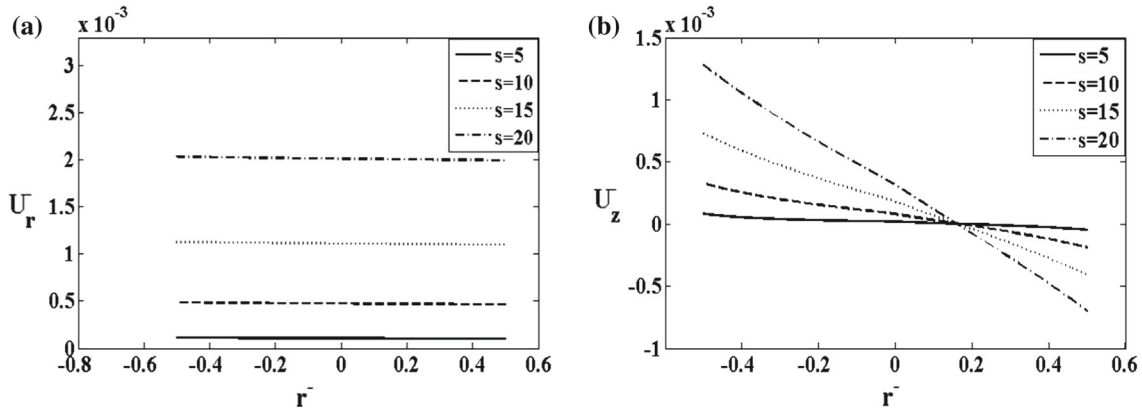


Fig. 6 Influence of mid-radius to thickness ratio on the dimensionless displacement and stress of the sandwich shell, CC boundary condition. **a** Radial displacement, **b** axial displacement

The state vector at the inner surface of the sandwich shell is computed by solving Eq. (21) and using relation (19a). Finally, the displacement and stress fields are determined by employing the obtained state vector in Eqs. (15a), (12) and (15b).

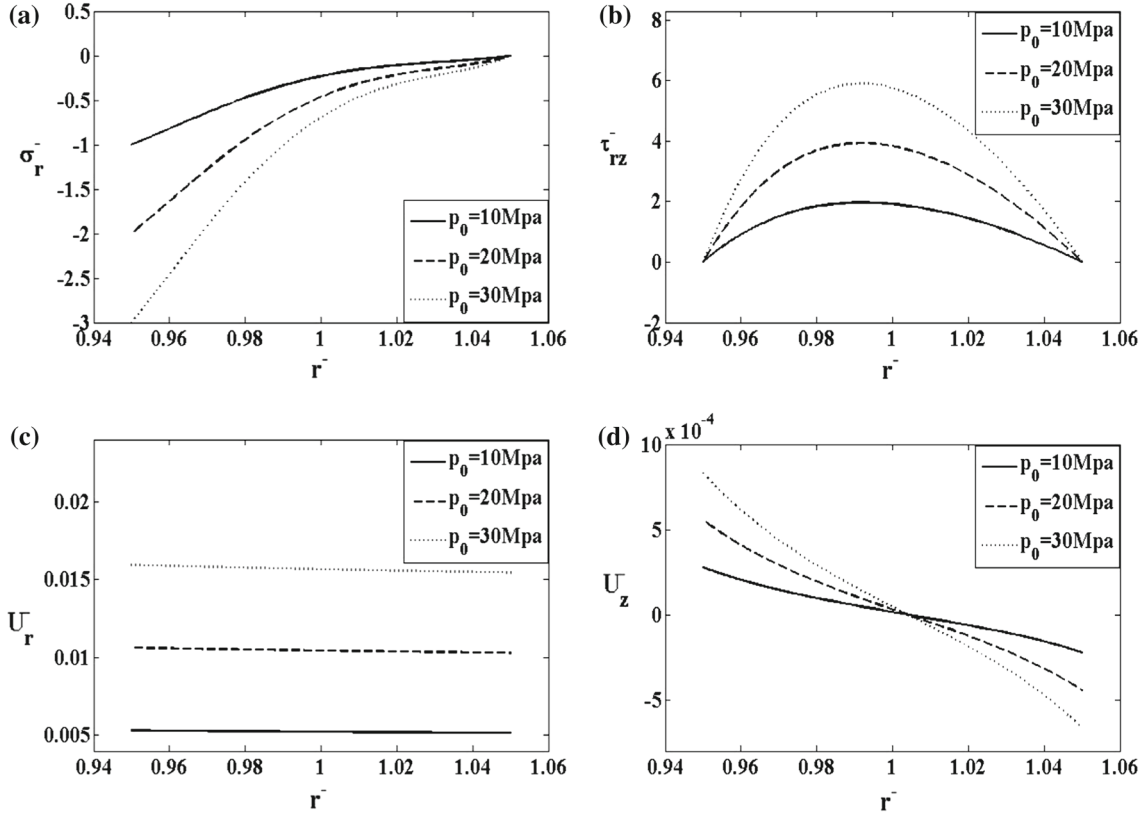


Fig. 7 Influence of mechanical load on the dimensionless stresses and displacements of the sandwich shell with CC boundary condition and $S = 10$. **a** Radial stress, **b** transverse shear stress, **c** radial displacement, **d** axial displacement

4 Semi-analytical solution

When the boundary condition is not simply supported, it is not possible to solve the governing differential equation analytically. We use DQM along the axial coordinate to solve the problem semi-analytically. This method approximates the n th partial derivatives of a function $f(x, z)$ with respect to x as follows:

$$\frac{\partial f^n(x, z)}{\partial x^n} \bigg|_{x=x_i} = \sum_{j=1}^N g_{ij}^{(n)} f(x_j, z), \quad (i = 1, 2, \dots, N; \quad n = 1, 2, \dots, N-1), \quad (22)$$

where $g_{ij}^{(n)}$ are the x_i -dependent weight coefficients and N is the number of sampling points in the x -domain. Sample points locations can be estimated as follows [27];

$$x_i = \frac{L}{2} \left(1 - \cos \frac{(i-1)\pi}{N-1} \right) \quad i = 1, 2, \dots, N. \quad (23)$$

Applying Eq. (22) to Eq. (11) results in the state-space equations at any sample point as follows:

$$\frac{d\Delta^-}{d\bar{r}} = M^- \Delta^-, \quad (24)$$

where $\Delta^- = \{\sigma_r^-, U_z^-, U_\theta^-, U_r^-, \tau_{rz}^-, \tau_{r\theta}^-\}$, $\sigma_r^- = \{\sigma_{r1}^-, \sigma_{r2}^-, \dots, \sigma_{rN}^-\}^T$ and M^- is introduced in Appendix IV.

Applying the edge boundary conditions (8a–c) into Eq. (24) leads to a new state-space equation as follows:

$$\frac{d\Delta_b^-}{d\bar{r}} = M_b^{-k} \Delta_b^- \quad (25)$$

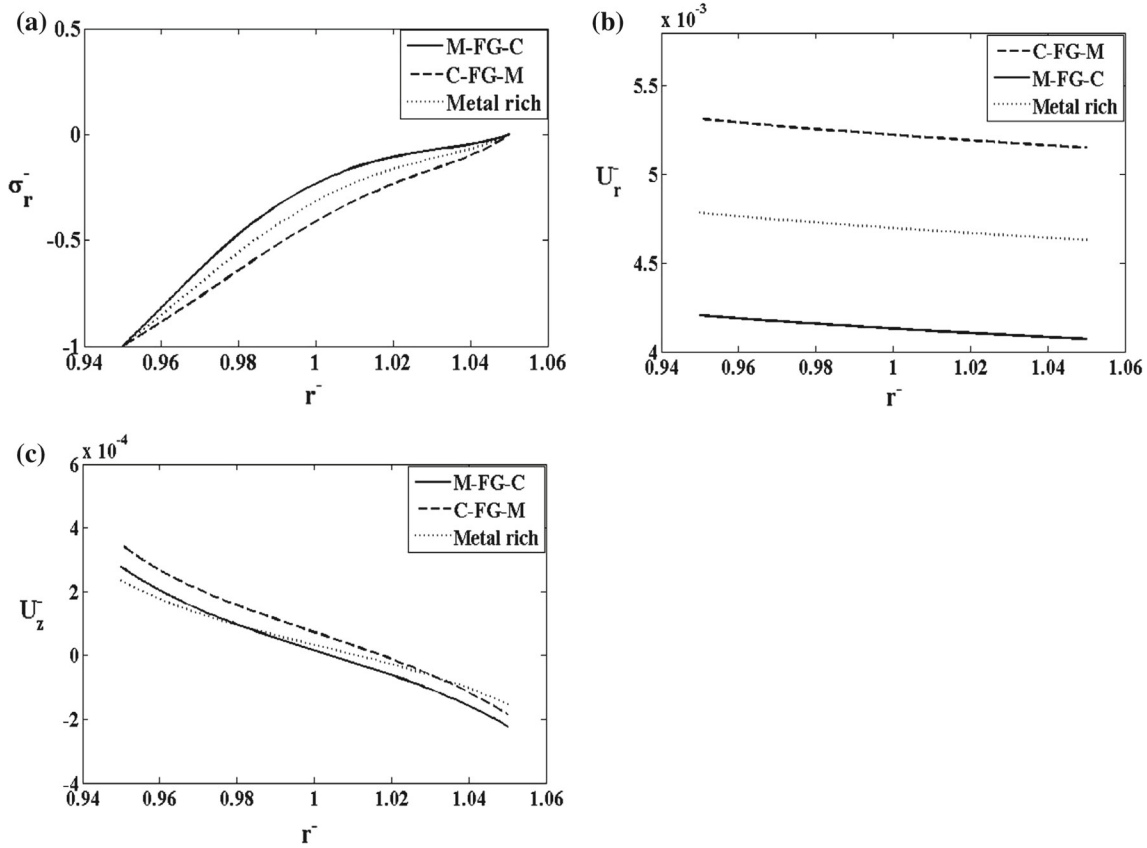


Fig. 8 Influence of graded direction on dimensionless stress and displacements sandwich shell with CC boundary condition and $S = 10$ **a** radial stress, **b** radial displacement, **c** axial stress

in which M_b^{-k} is defined for all boundary conditions in Appendix V.

Similarly, by solving Eq. (25) and using surface traction boundary conditions, the static and free vibration solution can be obtained.

5 Numerical results and discussion

A sandwich shell with monel/zirconia FGM core layer is considered to illustrate the present approach numerically. The material properties of the constituents and the geometric dimensions are as follows:

$$E_m = E_0 = 227.24 \text{ GPa}, \quad E_c = E_h = 125.83 \text{ GPa}, \quad \rho_m = \rho_0 = 227.24 \text{ GPa}, \quad \nu = 0.3, \\ \rho_c = \rho_h = 227.24 \text{ GPa}, \quad L = 3R_m, \quad s = R_m/h = 10, \quad h_{fg} = 20h_c, \quad p_0 = 10 \text{ MPa}, \quad h = 0.1.$$

5.1 Parametric study of static behavior

At first, convergence of the present approach for analytical and semi-analytical solutions is presented numerically. Numerical results for radial stress and displacement as well as transverse shear stress are computed and plotted in Fig. 2a–c. According to the figures, by increasing the half wave number along the axial coordinate up to 21, the through-the-thickness distribution of stresses and displacement converges to a constant value which demonstrates good convergence of the analytical solution. Numerical results for radial stress, radial displacement and transverse shear stress for a sandwich cylindrical shell with simply supported boundary conditions are computed by using DQM and presented in Fig. 3a–c. As concluded from this figure, when the number of sampling points reaches $N = 11$, the DQM solution converges to the analytical solution, which represents fast convergence of the semi-analytical approach. In order to validate the accuracy of the static solution, the

Table 2 Convergence of dimensionless natural frequency $\bar{\omega} = \omega \sqrt{\frac{R_m \rho_m h}{p_0}}$, for SS boundary condition

Sample point (N)	$\bar{\omega}_{11}$	$\bar{\omega}_{22}$	$\bar{\omega}_{33}$
5	0.097	0.901	2.134
7	0.105	0.955	2.167
9	0.114	0.960	2.179
11	0.1149	0.963	2.1815
13	0.115	0.963	2.182

Table 3 Natural frequencies (Hz) with SS

R/h	(n, m)	Method	P=0	P=1
200	(1, 2)	Analytical Eq. (20)	5.1521	5.2943
		Matsunaga [6]	5.1746	5.3020
100	(1, 2)	Analytical Eq. (20)	7.4464	7.6249
		Matsunaga [6]	7.4566	7.6398
50	(1, 1)	Analytical Eq. (20)	12.7425	13.0719
		Matsunaga [6]	12.7683	13.0828
25	(1, 1)	Analytical Eq. (20)	12.7691	13.0896
		Matsunaga [6]	12.7792	13.0940
20	(1, 1)	Analytical Eq. (20)	12.7766	13.0976
		Matsunaga [6]	12.7874	13.1024

non-dimensional radial stress and the deflection for FGM cylindrical shell with simply supported boundary conditions are computed and compared with the results of Tutuncu [3] as shown in Table 1. From comparison, some discrepancy is observed due to the Frobenius series as well as exponential function modeling of the FGM material properties which has been used by Tutuncu [3]. Moreover, a parametric study has been conducted to consider effects of edge boundary conditions, length to mid-radius ratio, mid-radius to thickness ratio, gradient index and mechanical loading on the bending behavior of the sandwich cylindrical shell. Figures 4a, b shows the effect of the edge boundary condition on radial stress and displacement of the sandwich shell. From Fig. 4a, it can be observed that the through-thickness distribution of radial stress, σ_r^- , in CC boundary conditions is maximum, whereas it is minimum for the case of SS boundary conditions. Also according to Fig. 4b and as expected, the distribution of the deflection along the thickness direction in CC boundary conditions is minimum; however, it is maximum for the SS boundary conditions. The influence of length to mid-radius ratio on stress and displacement components is depicted in Fig. 5a–d. Increasing the length to mid-radius ratio causes increasing displacement components as well as radial stress, whereas transverse shear stress decreases. The effect of mid-radius to thickness ratio on radial and axial displacement components is depicted in Fig. 6a, b. It can be seen that when the mid-radius to thickness ratio increases, the stiffness of the shell decreases and consequently radial and axial displacement increases. The effect of mechanical load on stresses and the displacements is shown in Fig. 7a–d. As expected, the displacement and stress components increase due to an increase in the mechanical load. From Fig. 7c, d, it is concluded that the radial displacement is independent of the radial coordinate, whereas the axial displacement varies nonlinearly along the radial coordinate. Besides, from Fig. 7d it is concluded that the neutral surface does not coincide with the mid surface due to the presence of the FGM core layer. Figures 8a–c depict the effect of the stacking sequence on the through-the-thickness distribution of the radial stress and displacement components. From the figures, it is concluded that the radial stress and displacements have the minimum value when the metal face sheet is the inner layer.

5.2 Parametric study of free vibration behavior

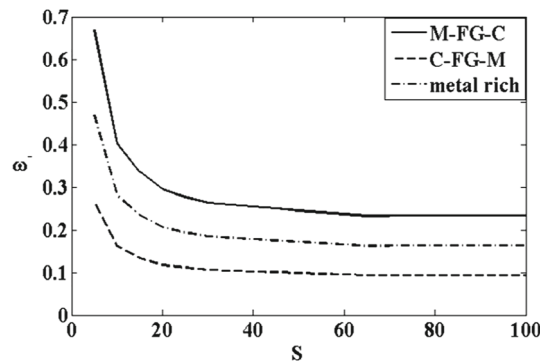
The natural frequencies for the sandwich cylindrical shell with the material properties and geometric dimension specified in the previous section are computed by solving Eq. (20). In order to show the convergence of the present technique, the dimensionless natural frequencies with different sample points are computed and tabulated in Table 2. According to the table by increasing the sampling points up to 13, the natural frequencies converge to constant values. Validation of present approach in vibration analysis is conducted by comparing the numerical results with related results in published literature. The natural frequencies for the FGM cylindrical shell with simply supported boundary condition are computed analytically and presented in Table 3. From comparison of the present results with the results reported by Matsunaga [6], almost good agreement can be

Table 4 Natural frequency (Hz) with CF ($L = 2$ m, $r_i = 0.95$ m, $h = 0.1$ m)

Mode	Method	$P = 0.6$	$P = 5$
1	DQM	151.13	147.32
	Su et al. [28]	152.26	148.08
2	DQM	151.13	147.32
	Su et al. [28]	152.26	148.07
3	DQM	218.91	217.45
	Su et al. [28]	219.67	218.32
4	DQM	218.91	217.45
	Su et al. [28]	219.66	218.32
5	DQM	252.28	243.81
	Su et al. [28]	253.36	243.99
6	DQM	252.28	243.81
	Su et al. [28]	253.35	243.99
7	DQM	383.58	381.74
	Su et al. [28]	384.21	382.02
8	DQM	383.58	381.74
	Su et al. [28]	384.22	382.01
9	DQM	417.65	403.34
	Su et al. [28]	418.12	404.10
10	DQM	426.63	418.99
	Su et al. [28]	418.12	419.27

Table 5 Dimensionless natural frequency for different conditions

(n,m)	SS	CC	CS	CF
(1,1)	0.336	0.402	0.379	0.305
(1,2)	0.566	0.612	0.588	0.517
(1,3)	1.112	1.191	1.168	0.981
(2,1)	0.963	0.996	0.987	0.915
(2,2)	1.424	1.481	1.446	1.410
(2,3)	2.187	2.207	2.195	2.153
(3,1)	1.697	1.732	1.706	1.623
(3,2)	2.182	2.218	2.201	2.147
(3,3)	2.833	2.889	2.867	2.790

**Fig. 9** Effect of mid-radius to thickness ratio on variation of dimensionless fundamental frequency with various stacking sequence of sandwich shell and CC boundary condition

achieved. Some discrepancy is observed due to the higher-order deformation theory used by Matsunaga [6]. Further comparison is made by comparing the present numerical results for the natural frequencies of the FGM cylindrical shell with clamped-free boundary conditions with the results reported by Su et al. [27] in Table 4. From comparison, good agreement, especially in higher modes, can be observed. The influence of edge boundary conditions on the natural frequencies is depicted in Table 5. According to the table and as expected, the natural frequencies increase by increasing the half wave number along the radial and circumferential direction, (n, m) . Moreover, it is concluded that the axial half wave number is more significant. As the table shows, irrespective of half wave numbers, the natural frequencies in CC boundary conditions are maximum, and they

are minimum for the CF boundary conditions. The influence of the mid-radius to thickness ratio, $S = \frac{R_m}{h}$, on the dimensionless fundamental frequency is presented in Fig. 9. Regardless of the stacking sequence, by increasing the mid-radius to thickness ratio the natural frequency decreases and when $S > 20$ the natural frequencies do not depend on S which denotes the thin shell behavior. In addition, as the figure shows in the case of M-FG-C stacking sequence, the natural frequency at each S is greater than that for the other two cases. So an important conclusion is that the sandwich cylindrical shell with inner and outer face sheets made by metal and ceramic, respectively, is a suitable stacking sequence from the viewpoint of vibration behavior.

6 Conclusion

- Radial stress in CC boundary conditions is maximum, whereas it is minimum for the case of SS boundary conditions, and for deflection it is vice versa.
- By increasing the mid-radius to thickness ratio, radial and axial displacement increases.
- The radial displacement is independent of the radial coordinate, whereas the axial displacement varies nonlinearly along the radial coordinate.
- The neutral surface does not coincide with the mid surface due to the presence of the FGM core layer.
- In the case of M-FG-C stacking sequence, the radial stress and displacement components are minimum and the natural frequencies are maximum compared to the other stacking sequences.
- The effect of the axial half wave number on the natural frequencies is more significant.
- The natural frequencies in CC boundary the conditions are maximum, and they are minimum for the CF boundary conditions.
- The natural frequency decreases by increasing the length to mid-radius ratio to $S = 20$, and then it is independent of S due to the thin shell behavior.

Appendix

Appendix I

$$G_f = \begin{bmatrix} \left(-\frac{1}{r} + \frac{v}{r(1-v)}\right) \frac{Ev}{r(1-v^2)} \frac{\partial}{\partial z} & \frac{E}{r^2(1-v^2)} \frac{\partial}{\partial \theta} & \left(\frac{E}{r^2(1-v^2)} + \rho \frac{\partial^2}{\partial t^2}\right) \frac{\partial}{\partial z} & -\frac{1}{r} \\ 0 & 0 & -\frac{\partial}{\partial z} & \frac{2(1+v)}{E} 0 \\ 0 & \frac{1}{r} & -\frac{1}{r} \frac{\partial}{\partial \theta} & \frac{2(1+v)}{E} \\ \frac{(1+v)(1-2v)}{(1-v)E} & -\frac{v}{1-v} \frac{\partial}{\partial z} & -\frac{v}{r(1-v)} \frac{\partial}{\partial \theta} & 0 \\ -\frac{v}{1-v} \frac{\partial}{\partial z} & -\left(\frac{E}{1-v^2} \frac{\partial^2}{\partial z^2} - \rho \frac{\partial^2}{\partial t^2}\right) & -\frac{v^2}{r(1-v)} \frac{\partial}{\partial \theta} & -\frac{Ev}{r(1-v^2)} \frac{\partial}{\partial z} \\ 0 & 0 & -\frac{E}{2(1+v)} \frac{\partial^2}{\partial z^2} + \rho \frac{\partial^2}{\partial t^2} & 0 \end{bmatrix}$$

Appendix II

$$G_f^- = \begin{bmatrix} a & \frac{E_0^- h v p_n^-}{r^- L (1-v^2)} & -\frac{E_0^- h \beta_m^-}{r^{-2} R_m (1-v^2)} & b & p_n^- \frac{R_m}{L} & \frac{\beta_m^-}{r^-} \\ 0 & 0 & 0 & -p_n^- \frac{R_m}{L} & \frac{2(1+v)}{E_0^-} \frac{R_m}{h} & 0 \\ 0 & 0 & \frac{1}{r^-} & \frac{\beta_m^-}{r^-} & 0 & \frac{2(1+v)}{E_0^-} \frac{R_m}{h} \\ c & \frac{R_m v}{1-v} \frac{p_n^-}{L} & \frac{v \beta_m^-}{r^- (1-v)} & -\frac{v}{r^- (1-v)} & 0 & 0 \\ -\frac{R_m v}{1-v} \frac{p_n^-}{L} & d & \frac{E_0^- p_n^- \beta_m^- h}{2 L r^- (1-v)} & -\frac{E_0^- p_n^- v h}{r^- L (1-v^2)} & -\left(\frac{m_1+1}{r^-}\right) & 0 \\ -\frac{\beta_m^- v}{r^- (1-v)} & \frac{E_0^- p_n^- \beta_m^- h}{2 L r^- (1-v)} & e & -\frac{h E_0^- \beta_m^-}{r^{-2} R_m (1-v^2)} & 0 & -\left(\frac{m_1+2}{r^-}\right) \end{bmatrix}$$

where

$$\begin{aligned}
 a &= \left(-\frac{1}{r^-} - \frac{m_1}{r^-} + \frac{v}{r^-(1-v)} \right), \quad b = \frac{E_0^- h}{r^{-2} R_m (1-v^2)} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1}, \\
 c &= \frac{(1+v)(1-2v)}{(1-v) E_0^-} \frac{R_m}{h}, \quad d = \frac{E_0^-}{1+v} \left(\frac{h R_m}{L^2} \frac{p_n^{-2}}{1-v} + \frac{h}{R_m} \frac{\beta_m^{-2}}{2r^-} \right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1}, \\
 e &= \frac{E_0^-}{1+v} \left(\frac{h R_m}{L^2} \frac{p_n^{-2}}{2(1+v)} + \frac{h}{R_m} \frac{\bar{\beta}_m^2}{r^{-2}(1-v)} \right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1}
 \end{aligned}$$

Appendix III

$$T = \begin{bmatrix} \left(\frac{r_o^- - h_c^-}{r_i^- + h_m^-} \right)^{m_1} & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \left(\frac{r_o^- - h_c^-}{r_i^- + h_m^-} \right)^{m_1} & 0 \\ 0 & 0 & 0 & 0 & 0 & \left(\frac{r_o^- - h_c^-}{r_i^- + h_m^-} \right)^{m_1} \end{bmatrix}$$

Appendix IV

$$M^- = \begin{bmatrix} a I_N & \frac{E^- h v}{r^-(1-v^2)} g_{ij}^- & -\frac{E^- h \beta_m^-}{r^{-2} R_m (1-v^2)} I_N & b I_N & -R_m g_{ij}^- & \frac{\beta_m^-}{\bar{r}} I_N & \frac{\beta_m^-}{\bar{r}} I_N \\ 0 & 0 & 0 & -R_m g_{ij}^- & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_N & 0 & 0 \\ 0 & 0 & \frac{1}{\bar{r}} I_N & \frac{\beta_m^-}{\bar{r}} I_N & 0 & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_N & 0 \\ c I_N & -\frac{R_m v}{1-v} g_{ij}^- & \frac{v \beta_m^-}{\bar{r}(1-v)} I_N & -\frac{v}{r^-(1-v)} I_N & 0 & 0 & 0 \\ -\frac{R_m v}{1-v} g_{ij}^- & d & \frac{E_0^- \beta_m^- h}{2\bar{r}(1-v)} g_{ij}^- & -\frac{E_0^- v h}{\bar{r}(1-v^2)} g_{ij}^- & -\left(\frac{m_1+1}{\bar{r}} \right) I_N & 0 & 0 \\ -\frac{\beta_m^- v}{\bar{r}(1-v)} I_N & \frac{E_0^- \beta_m^- h}{2\bar{r}(1-v)} g_{ij}^- & e & -\frac{h E_0^- \beta_m^-}{\bar{r}^2 R_m (1-v^2)} I_N & 0 & -\left(\frac{m_1+2}{\bar{r}} \right) I_N & 0 \end{bmatrix}$$

where

$$\begin{aligned}
 a &= \left(-\frac{1}{\bar{r}} - \frac{m_1}{\bar{r}} + \frac{v}{\bar{r}(1-v)} \right), \quad b = \frac{E^- h}{r^{-2} R_m (1-v^2)} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1}, \\
 c &= \frac{(1+v)(1-2v)}{(1-v) E^-} \frac{R_m}{h}, \quad d = \frac{E^-}{1+v} \left(\frac{h R_m}{1-v} g_{ij}^{-2} + \frac{h}{R_m} \frac{\beta_m^{-2}}{2r^-} I_N \right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1} I_N, \\
 e &= \frac{E^-}{1+v} \left(\frac{h R_m}{2(1+v)} g_{ij}^{-2} + \frac{h}{R_m} \frac{\beta_m^{-2}}{r^{-2}(1-v)} I_N \right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1} I_N
 \end{aligned}$$

Appendix V

Simply-simply

$$\Delta_b^- = \begin{bmatrix} aI_{N-2} & \frac{E^-hv}{r^-(1-v^2)}A_{ij}^- - \frac{E^-h\beta_m^-}{r^{-2}R_m(1-v^2)}I_{N-2} & bI_{N-2} & -R_mA_{ij}^- & \frac{\beta_m^-}{r^-}I_{N-2} \\ 0 & 0 & 0 & -R_mB_{ij}^- & \frac{2(1+v)}{E^-}\frac{R_m}{h}I_N & 0 \\ 0 & 0 & \frac{1}{r^-}I_{N-2} & \frac{\beta_m^-}{r^-}I_{N-2} & 0 & \frac{2(1+v)}{E^-}\frac{R_m}{h}I_{N-2} \\ cI_{N-2} & -\frac{R_mv}{1-v}A_{ij}^- & \frac{v\beta_m^-}{r^-(1-v)}I_{N-2} & -\frac{v}{r^-(1-v)}I_{N-2} & 0 & 0 \\ -\frac{R_mv}{1-v}\bar{B}_{ij} & d & \frac{E_0^-\beta_m^-h}{2r^-(1-v)}g_{ij}^- & -\frac{E_0^-vh}{r^-(1-v^2)}\bar{B}_{ij} & -\left(\frac{m_1+1}{r^-}\right)I_N & 0 \\ -\frac{\beta_m^-v}{r^-(1-v)}I_{N-2} & \frac{E_0^-\beta_m^-h}{2r^-(1-v)}g_{ij}^- & e & -\frac{hE_0^-\beta_m^-}{r^{-2}R_m(1-v^2)}I_{N-2} & 0 & -\left(\frac{m_1+2}{r^-}\right)I_{N-2} \end{bmatrix}$$

where

$$\begin{aligned} a &= \left(-\frac{1}{r^-} - \frac{m_1}{r^-} + \frac{v}{r^-(1-v)}\right), \quad b = \frac{E^-h}{r^{-2}R_m(1-v^2)} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1}, \\ c &= \frac{(1+v)(1-2v)}{(1-v)E^-} \frac{R_m}{h}, \quad d = \frac{E^-}{1+v} \left(\frac{hR_m}{1-v}G_{ij}^- + \frac{h}{R_m} \frac{\beta_m^{-2}}{2r^-}I_N\right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1} I_N, \\ e &= \frac{E^-}{1+v} \left(\frac{hR_m}{2(1+v)}C_{ij}^- + \frac{h}{R_m} \frac{\beta_m^{-2}}{r^{-2}(1-v)}I_{N-2}\right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1} I_{N-2} \\ A_{ij}^- &= g_{ij}^- \quad (i = 2, \dots, N-1; j = 1, \dots, N), \quad B_{ij}^- = g_{ij}^- \quad (i = 1, \dots, N; j = 2, \dots, N-1), \\ C_{ij}^- &= g_{ij}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1), \\ G_{ij}^- &= g_{i1}g_{1j}^- + g_{iN}g_{Nj}^- - g_{ij}^{-2} \quad (i = 1, \dots, N; j = 1, \dots, N) \end{aligned}$$

Clamped-clamped

$$\Delta_b^- = \begin{bmatrix} aI_{N-2} & \frac{E^-hv}{r^-(1-v^2)}A_{ij}^- - \frac{E^-h\beta_m^-}{r^{-2}R_m(1-v^2)}I_{N-2} & b & -R_mA_{ij}^- & \frac{\beta_m^-}{r^-}I_{N-2} \\ 0 & 0 & 0 & -R_mA_{ij}^- & \frac{2(1+v)}{E^-}\frac{R_m}{h}I_{N-2} & 0 \\ 0 & 0 & \frac{1}{r^-}I_{N-2} & \frac{\beta_m^-}{r^-}I_{N-2} & 0 & \frac{2(1+v)}{E^-}\frac{R_m}{h}I_{N-2} \\ cI_{N-2} & -\frac{R_mv}{1-v}A_{ij}^- & \frac{v\beta_m^-}{r^-(1-v)}I_{N-2} & -\frac{v}{r^-(1-v)}I_{N-2} & 0 & 0 \\ -\frac{R_mv}{1-v}A_{ij}^- & d & \frac{E_0^-\beta_m^-h}{2r^-(1-v)}g_{ij}^- & -\frac{E_0^-vh}{r^-(1-v^2)}A_{ij}^- & -\left(\frac{m_1+1}{r^-}\right)I_{N-2} & 0 \\ -\frac{\beta_m^-v}{r^-(1-v)}I_{N-2} & \frac{E_0^-\beta_m^-h}{2r^-(1-v)}g_{ij}^- & e & -\frac{hE_0^-\beta_m^-}{r^{-2}R_m(1-v^2)}I_{N-2} & 0 & -\left(\frac{m_1+2}{r^-}\right)I_{N-2} \end{bmatrix}$$

where

$$\begin{aligned} a &= \left(-\frac{1}{r^-} - \frac{m_1}{r^-} + \frac{v}{r^-(1-v)}\right), \quad b = \left(\frac{E^-h}{r^{-2}R_m(1-v^2)} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1}\right) I_{N-2} - \frac{E^-R_mh}{2(1+v)}G_{ij}^-, \\ c &= \frac{(1+v)(1-2v)}{(1-v)E^-} \frac{R_m}{h}, \quad d = \frac{E^-}{1+v} \left(\frac{hR_m}{1-v}g_{ij}^- + \frac{h}{R_m} \frac{\beta_m^{-2}}{2r^-}I_{N-2}\right) - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1} I_{N-2}, \\ e &= -\frac{E^-R_mh}{2(1+v)}A_{ij}^- - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-}\right)^{m_2-m_1} I_{N-2}, \quad A_{ij}^- = g_{ij}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1), \\ G_{ij}^- &= g_{i1}g_{1j}^- + g_{iN}g_{Nj}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1), \\ g_{ij}^{-2} &= g_{ij}^-g_{ij}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1) \end{aligned}$$

Clamped-simply

$$\Delta_b^- = \begin{bmatrix} aI_{N-2} & \frac{E^- h v}{r^-(1-v^2)} \bar{A}_{ij} - \frac{E^- h \beta_m^-}{r^{-2} R_m (1-v^2)} I_{N-2} & bI_{N-2} & -R_m \bar{A}_{ij} & \frac{\beta_m^-}{r^-} I_{N-2} \\ 0 & 0 & 0 & -R_m B_{ij}^- & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_{N-1} & 0 \\ 0 & 0 & \frac{1}{r^-} I_{N-2} & \frac{\beta_m^-}{r^-} I_{N-2} & 0 & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_{N-2} \\ cI_{N-2} & -\frac{R_m v}{1-v} A_{ij}^- & \frac{v \beta_m^-}{r^-(1-v)} I_{N-2} & -\frac{v}{r^-(1-v)} I_{N-2} & 0 & 0 \\ -\frac{R_m v}{1-v} B_{ij}^- & d & \frac{E_0^- \beta_m^- h}{2r^-(1-v)} \bar{g}_{ij} & -\frac{E_0^- v h}{r^-(1-v^2)} B_{ij}^- & -\left(\frac{m_1+1}{E^-}\right) I_{N-1} & 0 \\ -\frac{\beta_m^- v}{r^-(1-v)} I_{N-2} & \frac{E_0^- \beta_m^- h}{2r^-(1-v)} \bar{g}_{ij} & e & -\frac{h E_0^- \beta_m^-}{r^{-2} R_m (1-v^2)} I_{N-2} & 0 & -\left(\frac{m_1+2}{E^-}\right) I_{N-2} \end{bmatrix}$$

where

$$\begin{aligned} a &= \left(-\frac{1}{r^-} - \frac{m_1}{r^-} + \frac{v}{r^-(1-v)} \right), \quad b = \left(\frac{E^- h}{r^{-2} R_m (1-v^2)} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1} \right) I_{N-2} - \frac{E^- R_m h}{2(1+v)} G_{1ij}^-, \\ c &= \frac{(1+v)(1-2v)}{(1-v)E^-} \frac{R_m}{h}, \quad d = -\frac{E^- R_m h}{1-v^2} g_{ij}^{-2} - \frac{E^- R_m h v^2}{(1-v^2)(1-2v)} G_{2ij}^- \\ &\quad + \frac{E^- R_m h}{1-v^2} G_{ij}^{'-} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1} I_{N-1}, \\ e &= -\frac{E^- R_m h}{2(1+v)} g_{ij}^{-/2}, \\ A_{ij}^- &= g_{ij}^- \quad (i = 2, \dots, N-1; j = 1, \dots, N-1), \\ G_{2ij}^- &= g_{iN}^- g_{Nj}^- \quad (i = 1, \dots, N-1; j = 1, \dots, N-1), \\ G_{1ij}^- &= g_{iN}^- g_{Nj}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1), \\ G_{ij}^{'-} &= g_{i1}^- g_{1j}^- \quad (i = 1, \dots, N-1; j = 1, \dots, N-1), \\ B_{ij}^- &= g_{ij}^- \quad (i = 1, \dots, N-1; j = 2, \dots, N-1), \\ g_{ij}^{-/2} &= g_{ij}^- g_{ij}^- \quad (i = 2, \dots, N-1; j = 2, \dots, N-1), \\ g_{ij}^{-2} &= g_{ij}^- g_{ij}^- \quad (i = 1, \dots, N-1; j = 1, \dots, N-1), \end{aligned}$$

Clamped-free

$$\Delta_b^- = \begin{bmatrix} a & \frac{E^- h v}{r^-(1-v^2)} (A_{ij}^- - \frac{C_{ij}^-}{g_{11}^-}) - \frac{E^- h \beta_m^-}{r^{-2} R_m (1-v^2)} bI_{N-1} & -R_m A_{ij}^- & \frac{\beta_m^-}{r^-} I_{N-1} \\ 0 & 0 & 0 & -R_m g_{i1}^- & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_{N-2} & 0 \\ 0 & 0 & \frac{1}{r^-} I_{N-1} & \frac{\beta_m^-}{r^-} I_{N-1} & 0 & \frac{2(1+v)}{E^-} \frac{R_m}{h} I_{N-1} \\ c & -\frac{R_m v}{1-v} (A_{ij}^- - \frac{C_{ij}^-}{g_{11}^-}) & \frac{v \beta_m^-}{r^-(1-v)} I_{N-1} & -\frac{v}{r^-(1-v)} I_{N-1} & 0 & 0 \\ -\frac{R_m v}{1-v} (B_{ij}^- - \frac{g_{i1}^-}{g_{11}^-}) & d & \frac{E_0^- \beta_m^- h}{2r^-(1-v)} \bar{g}_{ij} & f & -\left(\frac{m_1+1}{r^-}\right) I_{N-2} & 0 \\ -\frac{\beta_m^- v}{r^-(1-v)} I_{N-1} & \frac{E_0^- \beta_m^- h}{2r^-(1-v)} \bar{g}_{ij} & e & -\frac{h E_0^- \beta_m^-}{r^{-2} R_m (1-v^2)} I_{N-1} & 0 & -\left(\frac{m_1+2}{r^-}\right) I_{N-1} \end{bmatrix}$$

where

$$\begin{aligned} a &= \left(-\frac{1}{r^-} - \frac{m_1}{r^-} + \frac{v}{r^-(1-v)} \right) I_{N-1} - \frac{v^2}{r^-(1-v)} \frac{g_{i1}^-}{g_{11}^-}, \\ b &= \frac{E^- h}{r^{-2} R_m (1-v^2)} I_{N-1} - \frac{E^- R_m h}{2(1+v)} D_{ij}^- - \frac{E^- h v^2}{r^{-2} R_m (1-v^2)} \frac{g_{i1}^-}{g_{11}^-} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2-m_1} I_{N-1}, \end{aligned}$$

$$c = \frac{(1 + \nu)(1 - 2\nu)}{(1 - \nu)E^-} \frac{R_m}{h} I_{N-1} - \frac{R_m \nu^2 (1 + \nu)}{E^- h (1 - \nu)} \frac{g_{i1}^-}{g_{11}^-},$$

$$d = -\frac{\nu}{r^-(1 - \nu)} I_{N-1} + \frac{\nu^2}{r^-(1 - \nu)} \frac{g_{i1}^-}{g_{11}^-} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2 - m_1} I_{N-1},$$

$$e = -\frac{E^- R_m h}{1 - \nu^2} \frac{g_{ij}^-}{g_{11}^-} + \frac{E^- R_m h}{(1 - \nu^2)} \frac{G_{ij}^-}{g_{11}^-} - \omega^{-2} \left(\frac{r^-}{r_i^- + h_m^-} \right)^{m_2 - m_1} I_{N-1},$$

$$f = -\frac{E^- \nu h}{r^-(1 - \nu^2)} \left(B_{ij}^- - \frac{g_{i1}^{-2}}{g_{11}^-} \right),$$

$$A_{ij}^- = g_{ij}^- \quad (i = 1, \dots, N - 1; j = 2, \dots, N - 1), \quad B_{ij}^- = g_{ij}^- \quad (i = 2, \dots, N - 1; j = 1, \dots, N - 1),$$

$$C_{ij}^- = g_{i1}^- g_{1j}^- \quad (i = 1, \dots, N - 1; j = 2, \dots, N - 1),$$

$$D_{ij}^- = g_{iN}^- g_{Nj}^- \quad (i = 1, \dots, N - 1; j = 1, \dots, N - 1), \quad g_{ij}^{-2} = g_{ij}^- g_{ij}^- \quad (i = 2, \dots, N - 1; j = 2, \dots, N - 1),$$

$$g_{ij}^{-/2} = g_{ij}^- g_{ij}^- \quad (i = 1, \dots, N - 1; j = 1, \dots, N - 1), \quad G_{ij}^- = g_{i1}^{-2} g_{1j}^- \quad (i = 2, \dots, N - 1; j = 2, \dots, N - 1)$$

References

- Varadan, T.K., Bhaskar, K.: Bending of laminated orthotropic cylindrical shells-an elasticity approach. *Compos. Struct.* **17**(2), 141–156 (1991)
- Kabir, H.R.H., Askar, H.: free vibration response of cylindrical panels with higher order shear deformation theory. *Int. J. Struct. Stab. Dyn.* **5**(3), 409–434 (2005)
- Tutuncu, N.: Stresses in thick-walled FGM cylinders with exponentially-varying properties. *Eng. Struct.* **29**(9), 2032–2035 (2007)
- Alibeigloo, A.: Static and vibration analysis of axi-symmetric angle-ply laminated cylindrical shell using state space differential quadrature method. *Int. J. Press. Vessels Pip.* **86**(11), 738–747 (2009)
- Zhao, X., Lee, Y.Y., Liew, K.M.: Thermoelastic and vibration analysis of functionally graded cylindrical shells. *Int. J. Mech. Sci.* **51**(9–10), 694–707 (2009)
- Matsunaga, H.: Free vibration and stability of functionally graded circular cylindrical shells according to a 2D higher-order deformation theory. *Compos. Struct.* **88**(4), 519–531 (2009)
- Brischetto, S.: Effect of the through-the- thickness temperature distribution on the response of layered and composite shells. *Int. J. Appl. Mech.* **1**(4), 581–605 (2009)
- Pradyumna, S., Bandyopadhyay, J.N.: Free vibration and buckling of functionally graded shell panels in thermal environments. *Int. J. Struct. Stab. Dyn.* **10**(5), 1031–1053 (2010)
- Alibeigloo, A., Nouri, V.: Static analysis of functionally graded cylindrical shell with piezoelectric layers using differential quadrature method. *Compos. Struct.* **92**(8), 1775–1785 (2010)
- Alibeigloo, A., Chen, W.Q.: Elasticity solution for an FGM cylindrical panel integrated with piezoelectric layers. *Eur. J. Mech. A/Solids* **29**, 714–723 (2010)
- Sepiani, H.A., Rastgoo, A., Ebrahimi, F., Ghorbanpour Arani, A.: Vibration and buckling analysis of two-layered functionally graded cylindrical shell. Considering the effects of transverse shear and rotary inertia. *Mater. Des.* **31**(3), 1063–1069 (2010)
- Malekzadeh, P., Heydarpour, Y.: Response of functionally graded cylindrical shells under moving thermo-mechanical loads. *Thin-Walled Struct.* **58**, 51–66 (2012)
- Rahmani, O., Khalili, S.M.R., Thomsen, O.T.: A high-order theory for the analysis of circular cylindrical composite sandwich shells with transversely compliant core subjected to external loads. *Compos. Struct.* **94**(7), 2129–2142 (2012)
- Alibeigloo, A., Kani, A.M., Pashaei, M.H.: Elasticity solution for the free vibration analysis of functionally graded cylindrical shell bonded to thin piezoelectric layers. *Int. J. Press. Vessels Pip.* **89**, 98–111 (2012)
- Lal, A., Singh, B.N., Kale, S.: Stochastic thermal postkling response of laminated composite cylindrical shell panel with system randomness. *Int. J. Appl. Mech.* **4**(1), 1250009 (2012)
- Alipour, M.M., Shariyat, M.: An elasticity-equilibrium-based zigzag theory for axisymmetric bending and stress analysis of the functionally graded circular sandwich plates, using a Maclaurin-type series solution. *Eur. J. Mech. A/Solids* **34**, 78–101 (2012)
- Wu, C.P., Kuo, C.H.: A unified formulation of PVD-based finite cylindrical layer methods for functionally graded material sandwich cylinders. *Appl. Math. Model.* **37**(3), 916–938 (2013)
- Sofiyev, A.H., Kuruoglu, N.: Buckling and vibration of shear deformable functionally graded orthotropic cylindrical shells under external pressures. *Thin-Walled Struct.* **78**, 121–130 (2014)
- Alibeigloo, A., Liew, K.M.: Free vibration analysis of sandwich cylindrical panel with functionally graded core using three-dimensional theory of elasticity. *Compos. Struct.* **113**, 23–30 (2014)
- Sun, X.S., Tan, V.B.C., Chen, Y., Tan, L.B., Jaiman, R.K., Tay, T.E.: Stress analysis of multi-layered hollow anisotropic composite cylindrical structures using the homogenization method. *Acta Mech.* **225**, 1649–1672 (2014)
- Sharma, J.N., Sharma, P.K., Mishra, K.C.: Analysis of free vibrations in axisymmetric functionally graded thermoelastic cylinders. *Acta Mech.* **225**, 1581–1594 (2014)

22. Plagianakos, T.S., Papadopoulos, E.G.: Coupled higher-order layerwise mechanics and finite element for cylindrical composite and sandwich shells with piezoelectric transducers. *Eur. J. Mech. A/Solids* **54**, 11–23 (2015)
23. Sburlati, R., Kashtalyan, M.: Elasticity analysis of sandwich pipes with functionally graded inter layers. *Eur. J. Mech. A/Solids* **59**, 232–241 (2016)
24. Kugler, S., Fotiu, P., Murin, J.: Thermo-elasticity in shell structures made of functionally graded materials. *Acta Mech.* **227**, 1307–1329 (2016)
25. Thomas, B., Roy, T.: Vibration analysis of functionally graded carbon nanotube-reinforced composite shell structures. *Acta Mech.* **227**, 581–599 (2016)
26. Norouzi, H., Alibeigloo, A.: Three dimensional static analysis of viscoelastic FGM cylindrical panel using state space differential quadrature method. *Eur. J. Mech. A/Solids* **61**, 254–266 (2017)
27. Shu, C., Richards, B.E.: Application of generalized differential quadrature to solve two- dimensional incompressible Navier-Stokes equations. *Int. J. Numer. Meth. Fluids* **15**, 791–798 (1992)
28. Su, Z., Jin, G., Ye, T.: Three-dimensional vibration analysis of thick functionally graded conical, cylindrical shell and annular plate structures with arbitrary elastic restraints. *Compos. Struct.* **118**, 432–447 (2014)